

Aggregation & Grouping

specific

Whenever you need to take any column into consideration and inside that col whatever the unique value you have take that as an important parameter & count some specific value for example for that part

Ex: Consider learner table with columns

learnerId	LearnerFirstName	LearnerLastName	LearnerPhone
	Rohan		
	Mohan		
	Chandru		
	Bharum		
	Banu		
	Karan		

query: Count of the num of students who joined the course via LinkedIn, Youtube, community.

①
Select Source of Joining, count(*) as num of students
③
from learners
④
Group By Source of Joining

② *
count(*)
↓
count of ~~non null~~ all value including null

specific.col: Source of Joining

these are the unique part

based on this part you may count of each

so you need to group based on source of joining column

Learner's EmailID	Learner's Employee Date	Selected Course	YOE	Source of Joining
		C	4	LinkedIn
		Python	4	Youtube
		Java	2	LinkedIn
		Python C++	1	Community
		Java	4	Youtube
		C++	6	Community

① First select the column name we want to work

② What you want to do

In this case, I need count of each

③ from which table

④ Then group it

o/p

LinkedIn	2
Youtube	2
Community	2

Important note:

① whatever parameter you are using after GROUP BY for which you want to do grouping that parameter should always be there after select •

② After grouped column in select, then it should be you can only use aggregate operators, and nothing else *

Eg.:

Select SourceofJoining, count(*) as num-of-students
from learners
Group by SourceofJoining

~~Select SourceofJoining, SelectedCourse, count(*) as num-of-students
from learners
Group by SourceofJoining~~

Select SourceofJoining, SelectedCourse, count(*)
from learners
Group by SourceofJoining, SelectedCourse

count of
non-null
value

Polp

Linkedin

c

1

Youtube

python

1

Linkedin

Java

community

c++

2

Youtube

Java

1

Query:

Corresponding to each course, how many students have enrolled

✓ ~~Select~~ selected Course, Count(*) from learners group Selected Course;

Query:

Corresponding to each course individual source of joining, give the max YOE

✓ select source of joining, MAX(YOE) from learners group source of joining;

Query: (check second sql)

Display the records of those learners who have joined the course via more than 1 source of joining

Order of Execution

from	→	select	→	group by	→	count	↓	having	↓	select
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Select SourceofJoining, count(*) AS num_of_students
From learners
GROUP BY SourceofJoining
HAVING num_of_students > 1

Important note:

① wants to apply filter on top of ~~group by~~
Aggregate function i.e. after group by should be

Having

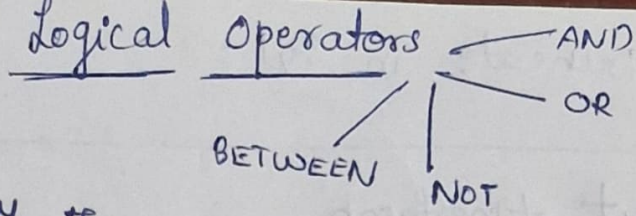
Query

Display the count of those students who joined via LinkedIn

Select SourceofJoining, count(*) as count
from learners

GROUP BY SourceofJoining

HAVING SourceofJoining = 'LinkedIn'



Query: Display the course which doesn't include ~~Excel~~ Excel

select * from course

where CourseName NOT LIKE "%EXCEL%".

ALTER (DDL Command) in SQL

ADD
MODIFY
DROP

Alter table employee ADD column jobdes varchar(50)

Alter table employee MODIFY column FirstName
already present
modify the column
varchar(50) to varchar(40)

to remove (x) drop ~~primary key~~ column

Alter table employee drop column jobdes

✓ If we try to drop primary key, the column should not be auto increment already.

✓ If any column is auto increment then it should be primary key / NOT NULL

-- Truncate in sql --

About timestamp

to record the time, whenever changes made in table,

✓ add column to note at creation

for eg:

Alter table learners add column

changed-At TIMESTAMP Default Now()

The change do not record timestamp, on update command

to use

changed-At Timestamp Default Now() on
update Now()